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Paper 1: A Systematic Review of Published Physiologically Based Pharmacokinetic Models fo

Bibliographic Information

- **Title:** A Systematic Review of Published Physiologically Based Pharmacokinetic Models for Drug Excretion Into Human Breastmilk: Knowledge Gaps and Opportunities to Optimize Reporting and Modeling Practices
- **Authors:** Kiptoo J, Okumu D, et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2026, 15(8)
- **DOI:** 10.1002/psp4.70321

Paper Type: GENERIC

Quick Take

This systematic review of 26 lactation PBPK studies (54 drugs) reveals major heterogeneity in reporting and modeling approaches, with only 18% of key reporting domains consistently documented. The authors propose a fit-for-purpose, prioritized reporting framework (LAC-PBPK) to improve reproducibility and regulatory confidence in PBPK-informed infant exposure assessment.

Executive Summary

Kiptoo et al. systematically reviewed 26 published PBPK studies modeling drug excretion into human breast milk and subsequent infant exposure, covering 54 medications. Using a novel 22-item abstraction tool (LAC-PBPK), they found that only 4 of 22 reporting domains (drug, mammary distribution model, model purpose, platform) were reported in all studies, with critical gaps in milk composition parameters, gestational age, transporter kinetics, and model evaluation criteria. Most studies used whole-body PBPK (88.5%) and assumed perfusion-rate-limited transfer (53%). The review identifies key modeling assumptions and limitations stemming from sparse clinical lactation data and physiological knowledge gaps, and proposes a prioritized, fit-for-purpose reporting framework distinguishing minimum essential, strongly recommended, and conditional elements. The authors argue PBPK retains value even when empirical data exist, by enabling mechanistic integration and scenario testing, but emphasize that transparent reporting and cautious selection of milk intake, evaluation, and extrapolation approaches are essential for reproducibility and regulatory confidence.

Scientific Context & Motivation

Breastfeeding is the preferred infant feeding modality, yet concerns about infant safety from maternal medications often lead to early cessation of breastfeeding. Regulatory guidance (ICH E21, EMA, FDA) increasingly recommends PBPK modeling to predict drug levels in breast milk and infant exposure. However, existing PBPK reporting guidance (EMA 2019, FDA 2018, Shebley

et al., Tan et al.) is generic or focused on environmental risk assessment, lacking lactation-specific structure. This creates heterogeneity in reporting and modeling approaches that impedes reproducibility, comparison, and confidence in model outputs. The review addresses the gap by systematically characterizing current practices and proposing a lactation-specific reporting framework.

✂ **Methodological Snapshot**

This is a systematic review following PRISMA guidelines, registered in PROSPERO (CRD42024607862). A literature search was conducted across Scopus, PubMed, and Google Scholar (through March 30, 2025) using keywords combining PBPK, human, breast milk, and lactation terms. Two independent reviewers screened articles, with a third adjudicating disagreements. Inclusion criteria were full-text articles using PBPK to evaluate plasma-to-milk transfer and infant exposure to human pharmaceuticals. Exclusion criteria included environmental chemicals, preclinical PBPK, and non-primary article types. The authors developed a 22-item data abstraction tool (LAC-PBPK) organized into three domains (Reporting, Data, Methodological), derived from existing regulatory and consortia guidelines (EMA, FDA, Shebley et al., Tan et al.). The tool was piloted on five studies with iterative refinement. Data were summarized using descriptive statistics (percentages) and narrative synthesis. No meta-analysis or formal risk-of-bias assessment was performed.

📋 **Detailed Analysis**

The review's core contribution is the LAC-PBPK framework and the empirical characterization of reporting practices. The finding that only 18% of reporting domains were consistently reported underscores a systemic issue in lactation PBPK: models are often developed with implicit assumptions that are not transparently documented. This is particularly concerning for regulatory applications where reproducibility is paramount. The heterogeneity in milk parameter sources (clinical vs. assumed vs. software defaults) and the wide ranges in milk pH (6.5–7.6) and creatinine (2%–12%) highlight the sensitivity of model outputs to these inputs, especially for weakly basic drugs subject to ion trapping. The observation that 61.5% of studies extrapolated beyond the observed infant age raises questions about the validity of predictions in unvalidated periods, particularly given the rapid developmental changes in infant physiology. The review's proposal of a fit-for-purpose framework is a pragmatic response to the field's maturity — it acknowledges that not all applications require the same evidentiary standard while still setting minimum expectations. The emphasis on milk intake estimation (fixed FDA estimate vs. population-based models) is well-placed, as this is a key driver of infant dose calculations. The limited use of infant plasma data (26.9%) is a notable gap, given that micro-sampling techniques exist. The review also correctly identifies the need for better data on mammary transporters, enzyme ontogeny, and preterm infant physiology. One subtle point: the review notes that only 3 studies modeled both parent drug and metabolite, which is important for drugs with active metabolites (e.g., codeine to morphine). The authors' argument that PBPK adds value beyond empirical data is well-reasoned — PBPK provides a mechanistic framework for scenario testing and extrapolation that empirical

data alone cannot. However, the review stops short of evaluating whether improved reporting actually improves model performance, which would be a valuable next step.

Domain Context

Lactation PBPK modeling sits at the intersection of maternal pharmacology, developmental pharmacology, and regulatory science. The field has grown rapidly since the first model in 2009 (Willman et al., codeine), driven by regulatory guidance (ICH E21, EMA, FDA) and the recognition that breastfeeding safety data are often lacking for new drugs. Key physiological considerations include: (1) postpartum changes in maternal physiology (renal filtration decreases ~2-fold in first 8 weeks), (2) milk composition dynamics (pH, creatocrit, protein content), (3) mammary drug transport (passive diffusion vs. active transporters), (4) infant developmental changes (enzyme ontogeny, renal maturation, gut permeability), and (5) milk intake variability. The field differs from environmental chemical lactation modeling in that pharmaceuticals require consideration of therapeutic dosing, maternal pharmacokinetics, and infant safety thresholds. Existing reporting guidelines (EMA, FDA) are generic and do not address lactation-specific elements like milk pH, creatocrit, mammary transporters, or milk intake estimation. The proposed LAC-PBPK framework addresses this gap. The review also highlights the tension between model complexity and data availability – while whole-body PBPK is the norm, the lack of lactation-specific physiological data forces reliance on assumptions. The field would benefit from standardized clinical lactation study protocols (e.g., from MILK-CENTRE, MPRINT) and open-source model sharing to accelerate progress.

☒ Key Findings

- 1) Only 4 of 22 LAC-PBPK reporting domains (drug modeled, mammary distribution model, model purpose, platform) were reported in all 26 studies; fewer than 50% reported breastmilk parameter inputs, gestational age, or transporter kinetics.
- 2) 88.5% of studies used whole-body PBPK; 53% assumed perfusion-rate-limited milk excretion; only 1 of 8 studies modeling potential transporter substrates included mammary transporter contribution.
- 3) 16 studies reported acceptance criteria, but only 13 prespecified both metric and value; only 7 (26.9%) used infant plasma data for evaluation; RID was the dominant exposure metric.
- 4) Simulated postpartum periods often did not match observed infant ages: 61.5% extrapolated beyond the clinical study age, and 11.5% had no overlap.
- 5) Milk pH and creatocrit were parameterized from heterogeneous sources (clinical, assumed, software defaults), with wide ranges.
- 6) Key modeling limitations stem from inadequate reporting of clinical lactation study attributes (feeding status, dosing history, sampling time) and empirical knowledge gaps (mammary transporter kinetics, preterm/early-infant physiology, enzyme ontogeny).
- 7) Sensitivity analyses identified drug (fraction unbound, lipophilicity, M/P ratio), maternal (breast permeability, milk properties), and infant (binding proteins, bioavailability, ontogeny) parameters as key drivers.
- 8) The review proposes a fit-for-purpose prioritization: minimum essential (maternal dose/formulation/route/history, postpartum timing, milk intake assumption, evaluation criteria), strongly recommended (milk pH/composition, gestational age/developmental physiology,

sensitivity analysis, infant plasma data), and conditional (transporter kinetics, parent-metabolite modeling).

Strengths & Limitations

Strengths

- Comprehensive systematic review following PRISMA guidelines with PROSPERO registration (CRD42024607862)
- Development and piloting of a novel 22-item LAC-PBPK data abstraction tool specifically for lactation PBPK reporting
- Broad coverage of 26 studies and 54 medications across multiple therapeutic classes
- Dual independent screening with adjudication by a third reviewer
- Detailed characterization of modeling approaches, assumptions, limitations, and sensitivity analyses
- Proposal of a practical, prioritized fit-for-purpose reporting framework (Table 2) that distinguishes minimum essential, strongly recommended, and conditional elements
- Clear articulation of the value of PBPK even when empirical data exist, emphasizing mechanistic integration and scenario testing

Limitations (Acknowledged by Authors)

- Only published journal articles included; unpublished studies and proprietary regulatory models not captured
- LAC-PBPK checklist may require future refinement for emerging modeling practices
- No formal risk-of-bias assessment due to lack of standardized appraisal tool for PBPK studies
- Review characterizes reporting and approaches, not predictive performance; findings should not be interpreted as validating any individual model

Limitations (Expert Review)

- Search limited to three databases (Scopus, PubMed, Google Scholar) and English-language articles, potentially missing relevant work in other databases or languages
- The LAC-PBPK tool was developed by the authors and piloted on only five studies; inter-rater reliability of the tool was not formally assessed
- Descriptive statistics without meta-analytic synthesis limit quantitative comparison of modeling approaches
- The review does not assess whether reporting completeness correlates with model predictive performance or regulatory acceptance
- The proposed fit-for-purpose framework is expert-opinion-based and not empirically validated against model outcomes
- Potential overlap in drug counts (54 medications across 26 studies) and the inclusion of multiple drugs per study may complicate interpretation of some percentages

Generalizability

The findings are generalizable to the published lactation PBPK literature for small-molecule pharmaceuticals, but may not fully represent proprietary regulatory submissions or models for biologics, environmental chemicals, or veterinary substances. The proposed framework is intended to be adaptable across compounds and populations, but its utility in regulatory contexts remains to be tested.

Figures & Tables

- **Figure 1:** PRISMA flow diagram showing study selection process, from initial search to 26 included studies.
 - *Significance:* Documents the systematic review methodology and transparency of study selection.
 - **Figure 2:** Bar chart showing the percentage of studies explicitly reporting each of the 22 LAC-PBPK reporting domains.
 - *Significance:* Highlights the key finding that only 4 of 22 domains were reported in all studies, illustrating widespread reporting gaps.
 - **Figure 3:** Summary of modeling features: (A) prespecification of performance acceptance criteria, (B) infant risk metric used (RID vs. plasma concentrations).
 - *Significance:* Illustrates heterogeneity in model evaluation approaches, with RID being the dominant metric and only a minority using infant plasma data.
 - **Figure 4:** Comparison of simulated postpartum ages versus observed infant ages from source clinical studies.
 - *Significance:* Demonstrates frequent extrapolation beyond observed data (61.5% of studies), raising concerns about model validity in extrapolated periods.
 - **Figure 5:** Summary workflow of PBPK modeling: key inputs, processes, assumptions, and metrics for evaluating breast milk secretion and infant exposure.
 - *Significance:* Provides a visual synthesis of the modeling framework and the key decision points in lactation PBPK.
 - **Figure 6:** Key considerations in advancing lactation PBPK modeling, including reporting frameworks, data generation, and open-source advocacy.
 - *Significance:* Summarizes the authors' recommendations for advancing the field.
 - **Table 1:** Characteristics of the 26 included studies: author, year, drug(s), route, purpose, model structure, maternal data, infant plasma data, acceptance criteria met, and software.
 - *Significance:* Provides a comprehensive overview of the included studies and their key modeling attributes, enabling cross-study comparison.
 - **Table 2:** Fit-for-purpose prioritization of lactation PBPK study-design and reporting elements, categorized as minimum essential, strongly recommended, or conditional/advanced.
 - *Significance:* The central deliverable of the review — a practical framework for improving reporting and study design in lactation PBPK.
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Code & Reproducibility Assessment

Not applicable — this is a systematic review, not a modeling study. The LAC-PBPK abstraction tool is provided as Table S1, but no analysis code or extracted dataset was made publicly available.

Supplementary Materials

Supplementary materials include Table S1 (LAC-PBPK data abstraction tool template), Table S2 (detailed study characteristics, assumptions, sensitivity analyses, and limitations), Figure S1 (data sources for model parameterization), Figure S2 (AI-drafted figure), Figure S3 (reported limitations), and the PRISMA checklist (Appendix).

Future Directions

1) Validate and refine the LAC-PBPK framework through application to a broader set of studies and regulatory submissions. 2) Develop standardized, validated modeling decision trees for lactation PBPK to guide sensitivity/uncertainty analyses and assumptions. 3) Generate empirical data on breastmilk intake for infants beyond 6 months and preterm populations (e.g., via IMI ConcePTION, MILK-CENTRE, MPRINT). 4) Fill knowledge gaps in mammary transporter kinetics, enzyme ontogeny (especially non-CYP and polymorphic systems), and milk composition dynamics. 5) Promote open-source PBPK platforms and software-agnostic model repositories to enhance transparency and reproducibility. 6) Compare infant exposure estimates from different milk intake quantification approaches. 7) Expand PBPK applications to postpartum-related diseases (e.g., gestational diabetes, hypertensive disorders). 8) Integrate transplacental exposure into postpartum risk assessment frameworks.

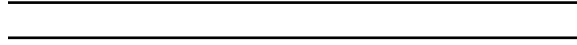
Expert Commentary

This review is a timely and valuable contribution to the lactation PBPK field, addressing a critical gap in reporting standards. The proposed fit-for-purpose framework is pragmatic and aligns with the maturity of the field, avoiding a one-size-fits-all mandate. The emphasis on distinguishing minimum essential from conditional elements is particularly useful for both academic and regulatory applications. However, the field would benefit from empirical validation of whether improved reporting actually translates to better model performance or regulatory acceptance. The call for open-source platforms and model repositories is well-founded, as reproducibility remains a major challenge in PBPK. From a pharmacometrics perspective, the review underscores the importance of transparently documenting assumptions about milk intake, physiological extrapolation, and parameter uncertainty — these are often the weakest links in lactation PBPK. The next step should be community consensus-building around the proposed framework, possibly through consortia like MPRINT or the IQ Consortium.

Bottom Line

Lactation PBPK modeling is rapidly adopted but suffers from inconsistent reporting and heterogeneous modeling assumptions. Researchers should adopt a fit-for-purpose reporting framework

(minimum essential, strongly recommended, conditional elements) to improve reproducibility and confidence. Clinicians and regulators should interpret PBPK-informed infant exposure predictions cautiously, particularly when milk intake assumptions, model evaluation criteria, and physiological extrapolations are not transparently documented.



☒ Figures

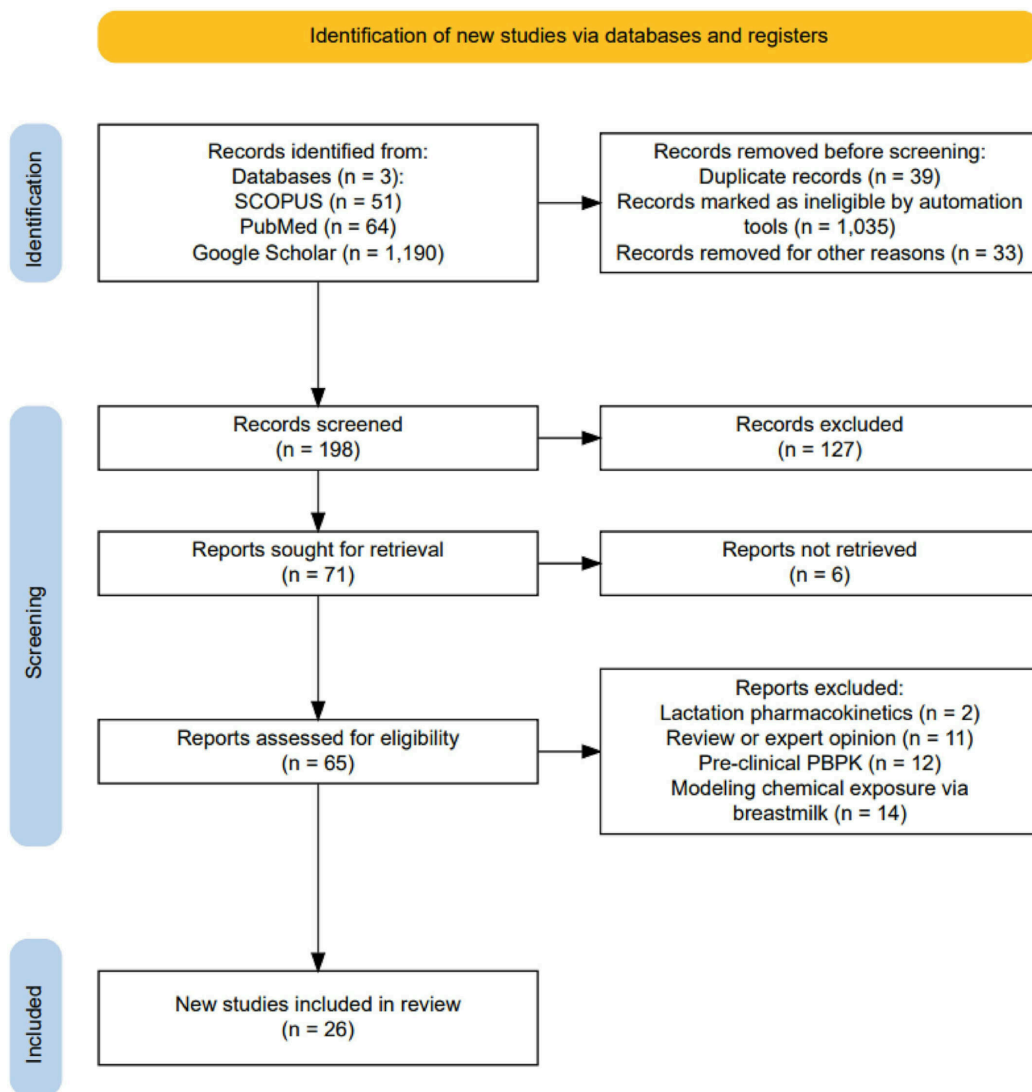


Figure 1: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 17 flow diagram of physiologically based pharmacokinetic (PBPK) study selection. Of

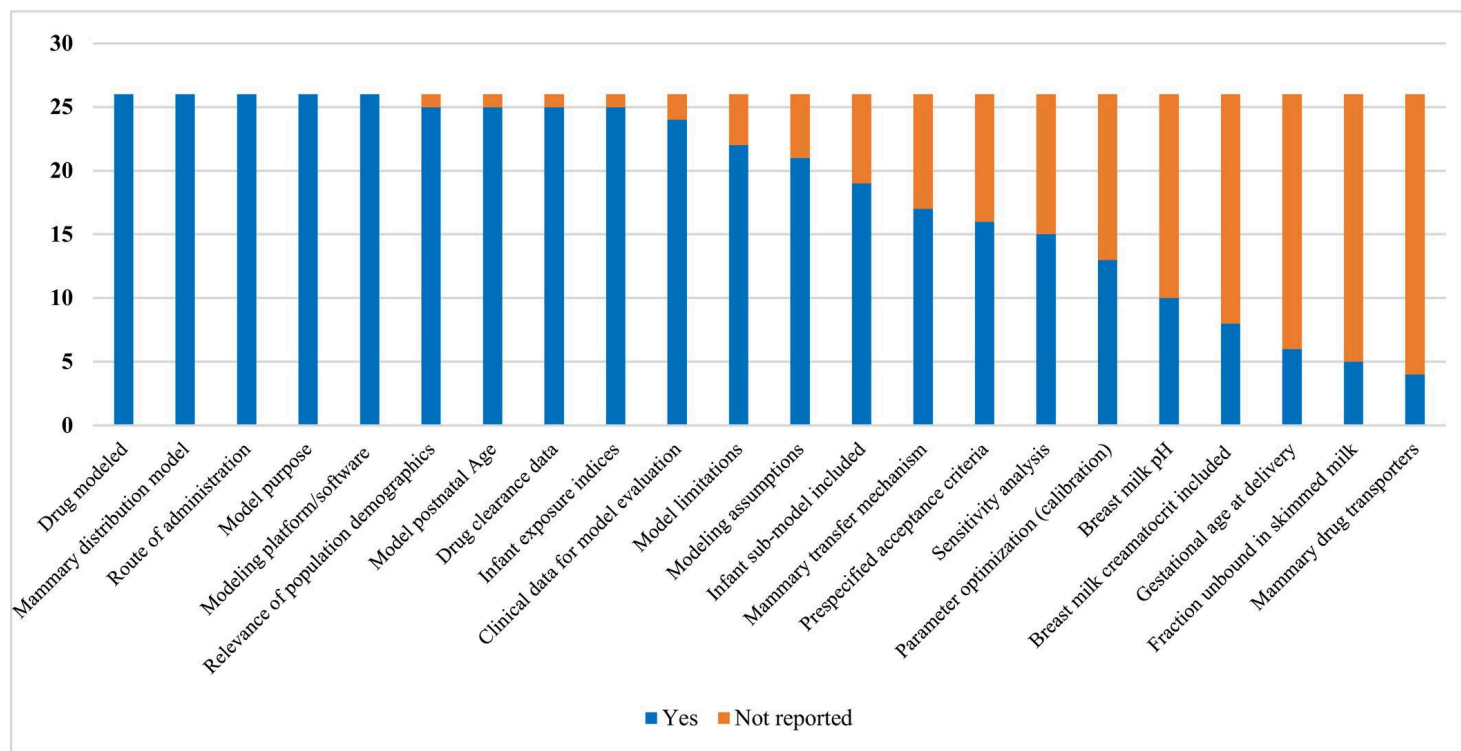


Figure 2: Reporting of PBPK model-related information in published papers. A bar chart to demonstrate descriptive summary “Yes/No” explicit reporting of PBPK model-related

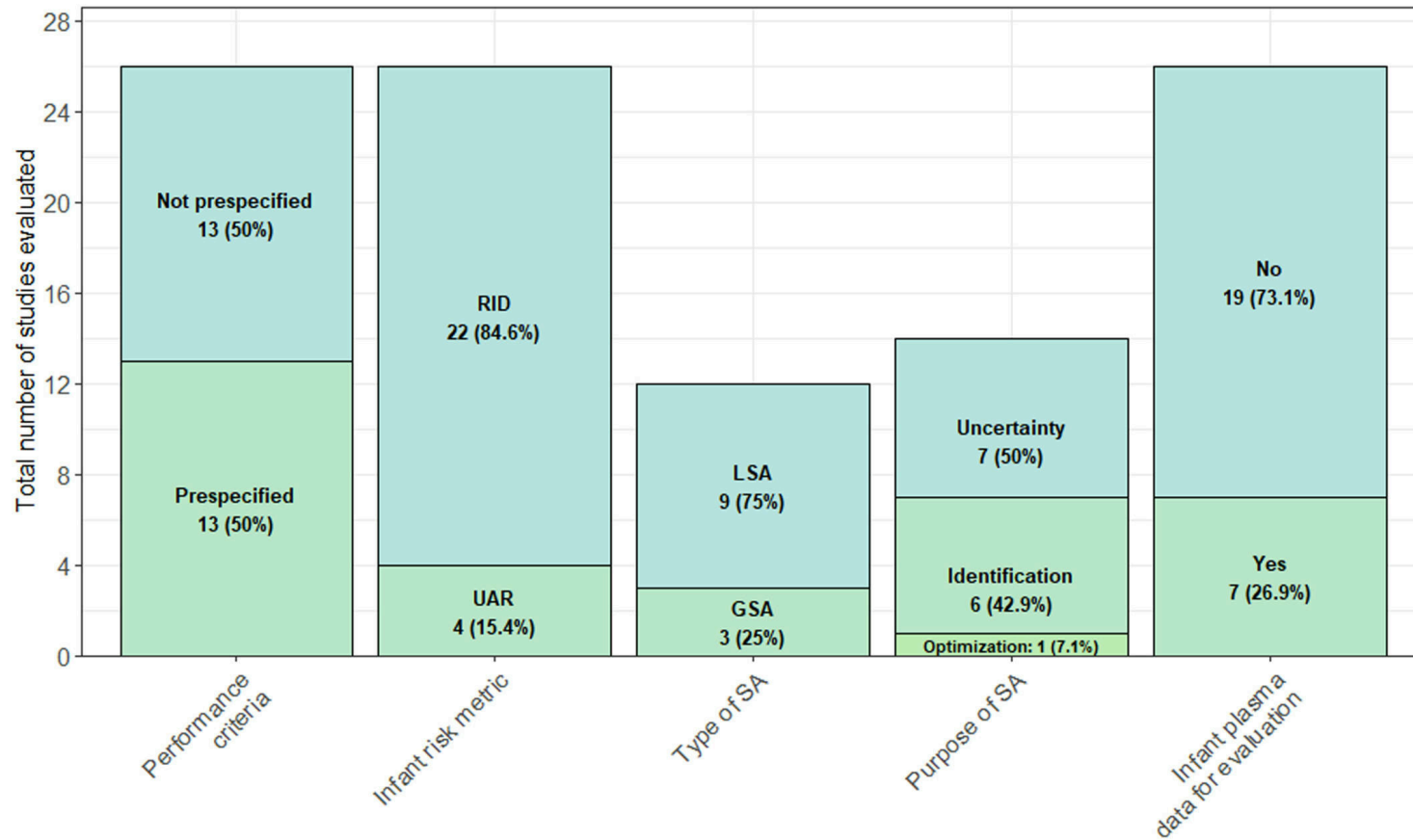


Figure 3: Summary of study modeling feature: (A) whether performance acceptance criteria were prespecified, (B) the infant risk metric used RID (relative infant dose) and

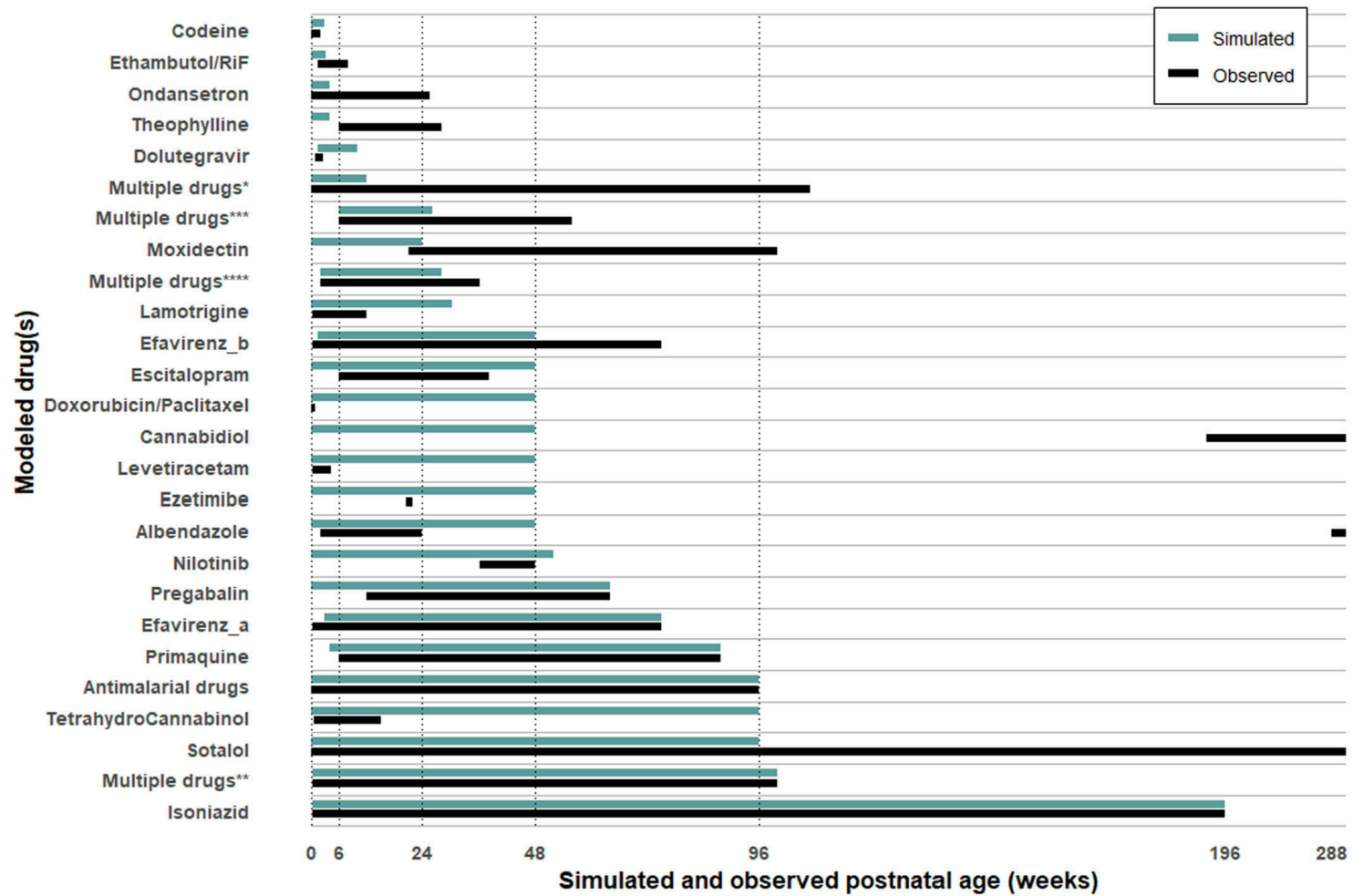


Figure 4: Simulated versus observed postnatal age. Postnatal ages from the source clinical studies (black) versus the ages used in the simulations (teal). Vertical dotted

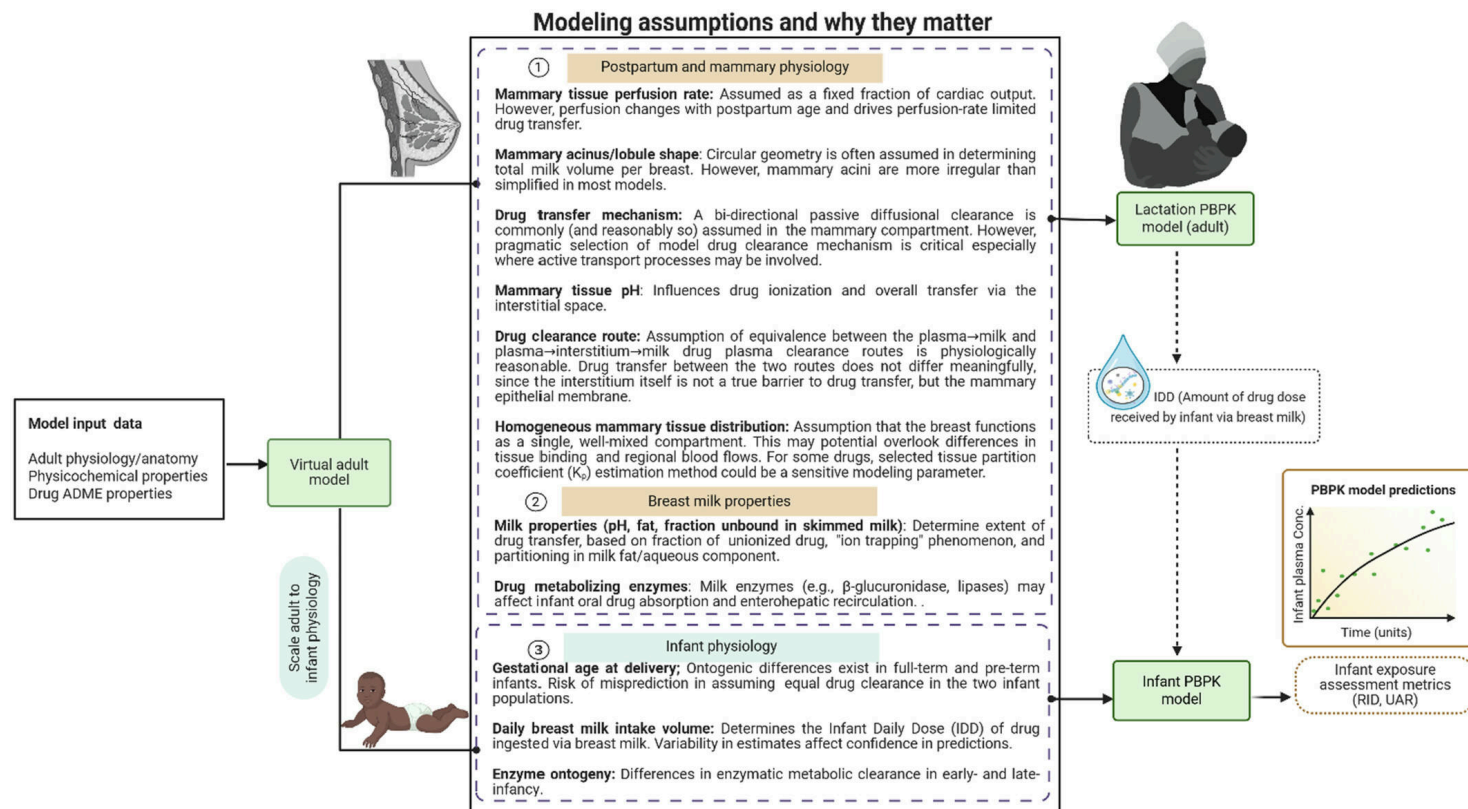


Figure 5: Summary workflow of PBPK modeling: Key modeling inputs, processes, assumptions, and metrics to evaluate breast milk secretion and infant exposure. ADME = absorpt

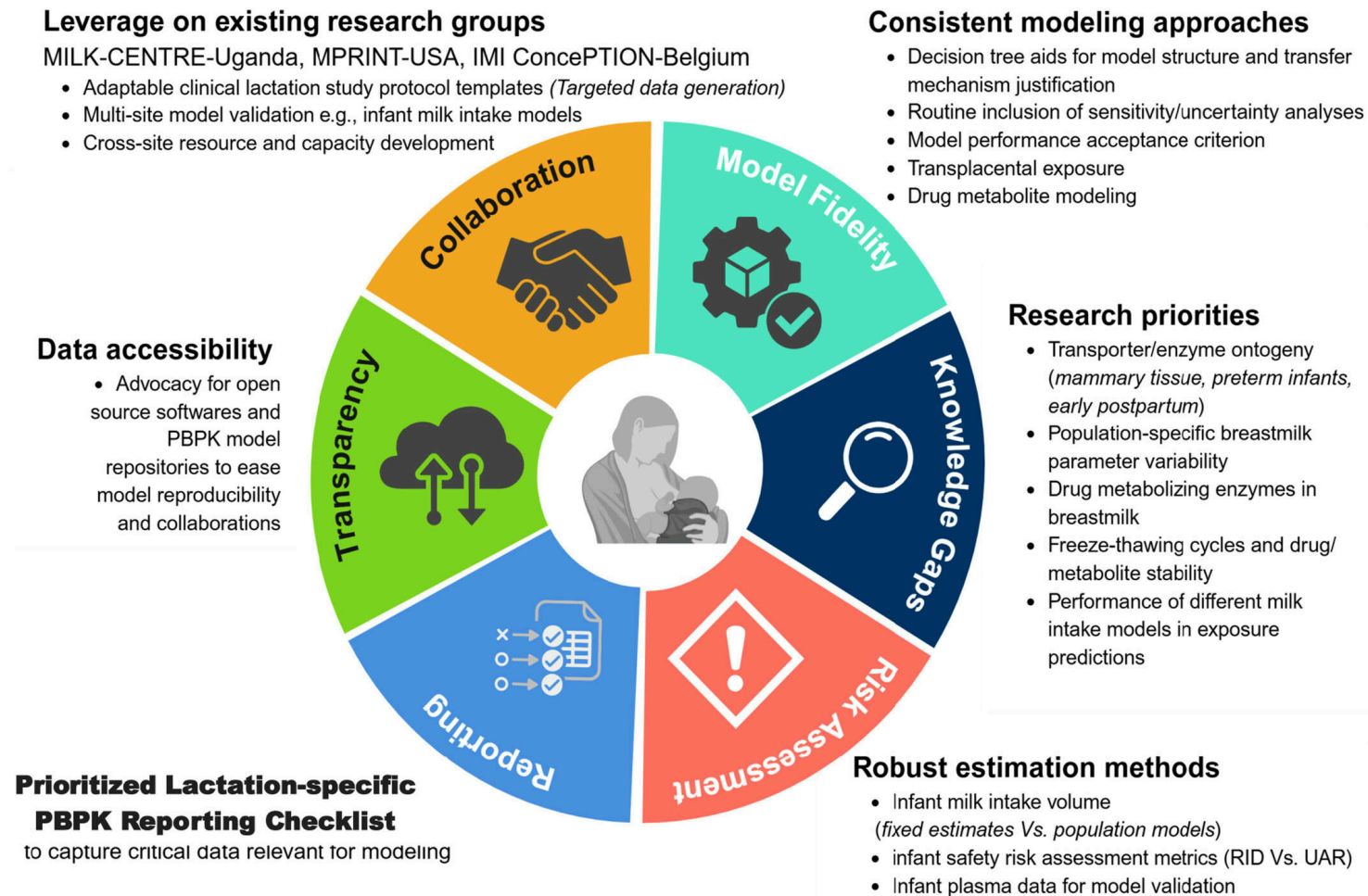


Figure 6: Key considerations in advancing Lactation PBPK modeling.